

Lie Groups, Lie Algebras, and Representations

Reading Notes and Exercises

Zhiyuan Xu
Department of Electrical and Computer Engineering
McGill University

Contents

1	Matrix Lie Groups	4
1.1	Definitions	4
1.2	Examples	4
1.2.1	General and Special Linear Groups	4
1.2.2	Unitary and Orthogonal Groups	4
1.2.3	Generalized Orthogonal and Lorentz Groups	5
1.2.4	Symplectic Groups	5
1.2.5	The Euclidean and Poincaré Groups	6
1.2.6	The Heisenberg Group	6
1.2.7	The Groups $\mathbb{R}^*$, $\mathbb{C}^*$, S^1 , $\mathbb{R}$, and $\mathbb{R}^n$	6
1.2.8	The Compact Symplectic Group	6
1.3	Topological Properties	7
1.3.1	Compactness	7
1.3.2	Connectedness	7
1.3.3	The Topology of $SO(3)$	8
1.4	Homomorphisms	8
1.5	Lie Groups	9
1.6	Exercises	9
2	The Matrix Exponential	11
2.1	The Exponential of a Matrix	11
2.2	Computing the Exponential	11
2.3	The Matrix Logarithm	12
2.4	Further Properties of the Exponential	13
2.5	The Polar Decomposition	13
2.6	Exercises	17
3	Lie Algebras	19
3.1	Definitions and First Examples	19
3.2	Simple, Solvable, and Nilpotent Lie Algebras	21
3.3	The Lie Algebra of a Matrix Lie Group	22
3.4	Examples	22
3.5	Lie Group and Lie Algebra Homomorphisms	22
3.6	The Complexification of a Real Lie Algebra	22
3.7	The Exponential Map	22
3.8	Consequences of Theorem 3.42	22
3.9	Exercises	22
4	Basic Representation Theory	24
4.1	Representations	24
4.2	Examples of Representations	25
4.3	New Representations from Old	26
4.3.1	Direct Sums	26
4.3.2	Tensor Products	26
4.3.3	Dual Representations	26
4.4	Complete Reducibility	26
4.5	Schur's Lemma	26
4.6	Representations of $\mathfrak{sl}(2; \mathbb{C})$	26
4.7	Group Versus Lie Algebra Representations	26
4.8	A Nonmatrix Lie Group	26

4.9 Exercises	26
-------------------------	----

1 Matrix Lie Groups

1.1 Definitions

Definition 1.1. The **general linear group** over the real numbers, denoted $GL(n, \mathbb{R})$, is the group of all $n \times n$ invertible matrices with real entries. The general linear group over the complex numbers, denoted $GL(n, \mathbb{C})$, is the group of all $n \times n$ invertible matrices with complex entries.

Definition 1.2. Let $M_n(\mathbb{C})$ denote the space of all $n \times n$ matrices with complex entries.

Definition 1.3. Let A_m be a sequence of complex matrices in $M_n(\mathbb{C})$. We say A_m **converges** to a matrix A if each entry of A_m converges to the corresponding entry of A as $m \rightarrow \infty$, i.e. if $(A_m)_{jk}$ converges to A_{jk} for all $1 \leq j, k \leq n$.

Note that Definition 1.3 means we can identify $M_n(\mathbb{C})$ with $\mathbb{C}^{n^2}$, and use the standard notion of convergence in $\mathbb{C}^{n^2}$. All topological properties of a matrix Lie group G will be considered with respect to the topology G inherits as a subset of $M_n(\mathbb{C}) \cong \mathbb{C}^{n^2}$.

We now consider subgroups of $GL(n, \mathbb{C})$, which are subsets $G \subset GL(n, \mathbb{C})$ such that $Id \in G$, and for all $A, B \in G$ we have $AB \in G$ and $A^{-1} \in G$.

Definition 1.4. A **matrix Lie group** is a subgroup G of $GL(n, \mathbb{C})$ with the property: if A_m is any sequence of matrices in G , and A_m converges to some matrix A , then either A is in G or A is not invertible.

This property means that G is a closed subset of $GL(n, \mathbb{C})$. So this definition is equivalent to saying a matrix Lie group is a closed subgroup of $GL(n, \mathbb{C})$. However, G is not necessarily closed in $M_n(\mathbb{C})$. For this to be true, we require a stronger property: if A_m is any sequence of matrices in G and A_m converges to some matrix A , then $A \in G$.

1.2 Examples

We introduce some important examples of (matrix) Lie groups, including the classical groups.

1.2.1 General and Special Linear Groups

The general linear groups $GL(n, \mathbb{R})$ and $GL(n, \mathbb{C})$, defined in 1.1, are matrix Lie groups.

The special linear groups $SL(n, \mathbb{R})$ and $SL(n, \mathbb{C})$ is the group of $n \times n$ invertible matrices (with real or complex entries) having determinant 1. They are matrix Lie groups.

1.2.2 Unitary and Orthogonal Groups

A $n \times n$ complex matrix A is **unitary** if its column vectors are orthonormal, that is,

$$\sum_{l=1}^n \bar{A}_{lj} A_{lk} = \delta_{jk}$$

where δ_{jk} is the Kronecker delta. This can be rewritten as

$$\sum_{l=1}^n (A^*)_{jl} A_{lk} = \delta_{jk}$$

where A^* is the adjoint of A , defined by $A^* = \bar{A}^t$. A is unitary if and only if $A^* = A^{-1}$.

The **unitary group** $U(n)$ is the collection of all unitary matrices. The **special unitary group** $SU(n)$ is the subgroup of $U(n)$ consisting of unitary matrices with determinant 1. $U(n)$ and $SU(n)$ are matrix Lie groups.

A $n \times n$ real matrix A is **orthogonal** if its column vectors are orthonormal. The equivalent conditions are the same as before, except that when A is real, $A^* = A^t$. A is orthogonal if and only if $A^t = A^{-1}$, if and only if A preserves the inner product on $\mathbb{R}^n$.

The **orthogonal group** $O(n)$ is the collection of all orthogonal matrices. The **special orthogonal group** $SO(n)$ is the set of $n \times n$ orthogonal matrices with determinant 1. Elements of $SO(n)$ are rotations, while the elements of $O(n)$ are either rotations or combinations of rotations and reflections.

Consider the bilinear form $(\cdot, \cdot)$ on $\mathbb{C}^n$ defined by

$$(x, y) = \sum_j x_j y_j$$

Note that this form is not an inner product, since it is symmetric, not conjugate-symmetric. The **complex orthogonal group** $O(n, \mathbb{C})$ is the set of all $n \times n$ complex matrices A that preserve this form. The group $SO(n, \mathbb{C})$ is the set of all matrices in $O(n, \mathbb{C})$ with determinant 1. $O(n, \mathbb{C})$ and $SO(n, \mathbb{C})$ are matrix Lie groups.

1.2.3 Generalized Orthogonal and Lorentz Groups

Consider $\mathbb{R}^{n+k}$, with n and k positive integers. Define a symmetric bilinear form $[\cdot, \cdot]_{n,k}$ on $\mathbb{R}^{n+k}$ by

$$[x, y]_{n,k} = x_1 y_1 + \cdots + x_n y_n - x_{n+1} y_{n+1} - \cdots - x_{n+k} y_{n+k}$$

The **generalized orthogonal group** $O(n, k)$ is the set of $(n+k) \times (n+k)$ real matrices A that preserve this form, i.e. $[Ax, Ay]_{n,k} = [x, y]_{n,k}$. It is a matrix Lie group. In particular, the **Lorentz group** is $O(3, 1)$. $SO(n, k)$ is the subgroup of $O(n, k)$ consisting of elements with determinant 1.

For any $A \in O(n, k)$, we have $\det A = \pm 1$.

1.2.4 Symplectic Groups

Define the skew-symmetric bilinear form ω on $\mathbb{R}^{2n}$ by

$$\omega(x, y) = \sum_{j=1}^n (x_j y_{n+j} - x_{n+j} y_j)$$

The **real symplectic group** $Sp(n, \mathbb{R})$ is the set of all $2n \times 2n$ real matrices A which preserve ω . It is a closed subgroup of $GL(2n, \mathbb{R})$, hence a matrix Lie group. $\det A = \pm 1$ for all $A \in Sp(n, \mathbb{R})$; in fact, $\det A = 1$.

Using the same formula, we can define a bilinear form ω on $\mathbb{C}^{2n}$. Over $\mathbb{C}$, we have the relation $\omega(z, w) = (z, \Omega w)$, where $(\cdot, \cdot)$ is the complex bilinear form from 1.2.2. The **complex symplectic group** $Sp(n, \mathbb{C})$ is the set of all $2n \times 2n$ complex matrices which preserve this form. We can show that for all $A \in Sp(n, \mathbb{C})$, $\det A = \pm 1$, and in fact, $\det A = 1$.

The **compact symplectic group** $Sp(n)$ is defined as

$$Sp(n) = Sp(n, \mathbb{C}) \cap U(2n)$$

That is to say, $Sp(n)$ is the group of $2n \times 2n$ matrices that preserve both the inner product and the bilinear form ω .

1.2.5 The Euclidean and Poincaré Groups

The **Euclidean group** $E(n)$ is the group of all transformations of $\mathbb{R}^n$ that can be expressed as a composition of a translation and an orthogonal linear transformation. Elements of $E(n)$ are written as pairs $\{x, R\}$ with $x \in \mathbb{R}^n$ and $R \in O(n)$, and $\{x, R\}$ acts on $\mathbb{R}^n$ by

$$\{x, R\}y = Ry + x$$

The product operation for $E(n)$ is

$$\{x_1, R_1\}\{x_2, R_2\} = \{x_1 + R_1x_2, R_1R_2\}$$

The inverse of an element of $E(n)$ is

$$\{x, R\}^{-1} = \{-R^{-1}x, R^{-1}\}$$

$E(n)$ is not a subgroup of $GL(n, \mathbb{R})$, but it is isomorphic to a closed subgroup of $GL(n+1, \mathbb{R})$.

The **Poincaré group** $P(n, 1)$ is the group of all transformations of $\mathbb{R}^{n+1}$ of the form

$$T = T_x A$$

with $x \in \mathbb{R}^{n+1}$ and $A \in O(n, 1)$.

1.2.6 The Heisenberg Group

The **Heisenberg group** H is the set of all 3×3 real matrices

$$A = \begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix}$$

where $a, b, c \in \mathbb{R}$. H is a subgroup of $GL(3, \mathbb{R})$.

1.2.7 The Groups $\mathbb{R}^*$, $\mathbb{C}^*$, S^1 , $\mathbb{R}$, and $\mathbb{R}^n$

These groups are not defined as groups of matrices, but can be thought of as such.

- The group $\mathbb{R}^*$ of non-zero real numbers under multiplication is isomorphic to $GL(1, \mathbb{R})$
- The group $\mathbb{C}^*$ of non-zero complex numbers under multiplication is isomorphic to $GL(1, \mathbb{C})$
- The group S^1 of complex numbers with absolute value 1 is isomorphic to $U(1)$
- The group $\mathbb{R}$ under addition is isomorphic to $GL(1, \mathbb{R})^+$, the 1×1 matrices with positive determinant
- The group $\mathbb{R}^n$ under vector addition is isomorphic to the group of diagonal real matrices with positive diagonal entries

1.2.8 The Compact Symplectic Group

We try to understand the structure of $Sp(n)$, and show that it can be understood as the "unitary group over the quaternions".

Define a conjugate-linear map $J : \mathbb{C}^{2n} \rightarrow \mathbb{C}^{2n}$ by

$$J(\alpha, \beta) = (-\bar{\beta}, \bar{\alpha})$$

where $\alpha, \beta \in \mathbb{C}^n$ and $(\alpha, \beta) \in \mathbb{C}^{2n}$. For all $z, w \in \mathbb{C}^{2n}$, we have

$$\omega(z, w) = \langle Jz, w \rangle$$

Furthermore, for all $z, w \in \mathbb{C}^{2n}$, we have

$$\langle Jz, w \rangle = -\langle Jw, z \rangle$$

and

$$J^2 = -I$$

Proposition 1.5. *If $U \in U(2n)$, then $U \in Sp(n)$ if and only if U commutes with J .*

Theorem 1.6. *$U \in Sp(n)$ if and only if there exists an orthonormal basis $u_1, \dots, u_n, v_1, \dots, v_n$ for $\mathbb{C}^{2n}$ such that the following properties hold:*

1. $Ju_j = v_j$
2. For some real numbers $\theta_1, \dots, \theta_n$, we have

$$\begin{aligned} Uu_j &= e^{i\theta_j}u_j \\ Uv_j &= e^{-i\theta_j}v_j \end{aligned}$$

3. We have

$$\begin{aligned} \omega(u_j, u_k) &= \omega(v_j, v_k) = 0 \\ \omega(u_j, v_k) &= \delta_{jk} \end{aligned}$$

Lemma 1.7. *Suppose V is a complex subspace of $\mathbb{C}^{2n}$ that is invariant under the conjugate-linear map J . Then the orthogonal complement $V^\perp$ of V (with respect to the inner product $\langle \cdot, \cdot \rangle$) is also invariant under J . Furthermore, V and $V^\perp$ are orthogonal with respect to ω , i.e.*

$$\omega(z, w) = 0$$

for all $z \in V$ and $w \in V^\perp$.

1.3 Topological Properties

1.3.1 Compactness

Definition 1.8. A matrix Lie group $G \subset GL(n, \mathbb{C})$ is **compact** if it is compact in the usual topological sense as a subset of $M_n(\mathbb{C}) \cong \mathbb{R}^{2n^2}$.

By the Heine-Borel theorem, a matrix Lie group G is compact if and only if it is closed and bounded.

$O(n)$, $SO(n)$, $U(n)$, $SU(n)$, and $Sp(n)$ are compact.

1.3.2 Connectedness

Definition 1.9. A matrix Lie group G is **connected** if for all $A, B \in G$, there exists a continuous path $A(t)$, $a \leq t \leq b$, lying in G with $A(a) = A$ and $A(b) = B$. For any matrix Lie group G , the **identity component** of G , denoted G_0 , is the set of $A \in G$ for which there exists a continuous path $A(t)$, $a \leq t \leq b$, lying in G with $A(a) = I$ and $A(b) = A$.

This definition is actually called **path connected** in topology. A matrix Lie group is connected if and only if it is path-connected.

To show that a matrix Lie group G is connected, it suffices to show that each $A \in G$ can be connected to the identity by a continuous path lying in G .

Proposition 1.10. *If G is a matrix Lie group, the identity component G_0 of G is a normal subgroup of G .*

G_0 is closed, and hence a matrix Lie group.

Proposition 1.11. *$GL(n, \mathbb{C})$ is connected for all $n \geq 1$.*

Proposition 1.12. *$SL(n, \mathbb{C})$ is connected for all $n \geq 1$.*

Proposition 1.13. *$U(n)$, $SU(n)$, and $SO(n)$ are connected for all $n \geq 1$.*

1.3.3 The Topology of $SO(3)$

Definition 1.14. *The **real projective space** of dimension n , denoted $\mathbb{R}P^n$, is the set of lines through the origin in $\mathbb{R}^{n+1}$. Since each line through the origin intersects the unit sphere exactly twice, we may think of $\mathbb{R}P^n$ as the unit sphere S^n with "antipodal" points u and $-u$ identified.*

Points in $\mathbb{R}P^n$ can be thought of as pairs $\{u, -u\}$ with $u \in S^n$. There is a natural map $\pi : S^n \rightarrow \mathbb{R}P^n$

$$\pi(u) = \{u, -u\}$$

A distance function can be defined on $\mathbb{R}P^n$ by

$$d(\{u, -u\}, \{v, -v\}) = \min(d(u, v), d(u, -v))$$

The space $\mathbb{R}P^n$ is the upper hemisphere in S^n with antipodal points on the equator identified. $\mathbb{R}P^n$ is the closed unit ball $B^n \subset \mathbb{R}^n$, with antipodal points on the boundary of B^n identified.

Proposition 1.15. *There is a continuous bijection between $SO(3)$ and $\mathbb{R}P^3$.*

$\mathbb{R}P^n$ is not simply connected, and hence $SO(3)$ is not simply connected.

1.4 Homomorphisms

Definition 1.16. *Let G and H be matrix Lie groups. A map Φ from G to H is called a **Lie group homomorphism** if*

1. Φ is a group homomorphism, and
2. Φ is continuous.

*If, in addition, Φ is one-to-one and onto and the inverse map Φ^{-1} is continuous, then Φ is called a **Lie group isomorphism**.*

If G and H are matrix Lie groups, and there exists a Lie group isomorphism from G to H , then G and H are isomorphic, $G \cong H$.

The relationship between $SU(2)$ and $SO(3)$: they are not isomorphic, but there exists a Lie group homomorphism $\Phi : SU(2) \rightarrow SO(3)$ that is two-to-one and onto.

Proposition 1.17. *The map $U \mapsto \Phi_U$ is a 2-1 and onto map of $SU(2)$ to $SO(3)$, with kernel equal to $\{I, -I\}$.*

$SU(2)$ is homeomorphic to S^3 . The proposition shows that $SO(3)$ is homeomorphic to $\mathbb{R}P^3$, that is, S^3 with antipodal points identified.

1.5 Lie Groups

The text first gives a very brief discussion of manifolds and smooth manifolds.

Definition 1.18. A **Lie group** is a smooth manifold G which is also a group, and such that the group product $G \times G \rightarrow G$ and the inverse map $G \rightarrow G$ are smooth.

Every matrix Lie group is a Lie group, but the reverse is not true. However, we focus on matrix Lie groups.

Usually, we call a map Φ between two Lie groups a Lie group homomorphism if Φ is a group homomorphism and Φ is smooth.

Every matrix Lie group G is a manifold, so G must be locally path connected, so G is connected if and only if it is path connected.

1.6 Exercises

Exercise 1.7 Using Theorem 1.6, show that $Sp(n)$ is connected and that every element of $Sp(n)$ has determinant 1.

Proof. Let $U \in Sp(n)$. Note that the matrix U represents a linear transformation $L_u : \mathbb{C}^{2n} \rightarrow \mathbb{C}^{2n}$. By Theorem 1.6, there exists an orthonormal basis $u_1, \dots, u_n, v_1, \dots, v_n$ for $\mathbb{C}^{2n}$ such that

1. $Ju_j = v_j$

2. for some $\theta_1, \dots, \theta_n \in \mathbb{R}$,

$$Uu_j = e^{i\theta_j}u_j$$

$$Uv_j = e^{-i\theta_j}v_j$$

3. and we have

$$\omega(u_j, u_k) = \omega(v_j, v_k) = 0$$

$$\omega(u_j, v_k) = \delta_{jk}$$

The second point is the most important. Notice that it implies $e^{i\theta_j}$ are eigenvalues of U with eigenvectors u_j , and $e^{-i\theta_j}$ are eigenvalues of U with eigenvectors v_j .

Denote $B = \{u_1, \dots, u_n, v_1, \dots, v_n\}$. B is a basis for $\mathbb{C}^{2n}$ consisting entirely of eigenvectors of U , so U is diagonalizable. Thus there exists an invertible matrix $Q \in M_{2n}(\mathbb{C})$ such that

$$QUQ^{-1} = \text{diag}(e^{i\theta_1}, \dots, e^{i\theta_n}, e^{-i\theta_1}, \dots, e^{-i\theta_n})$$

and so

$$U = Q^{-1} \text{diag}(e^{i\theta_1}, \dots, e^{i\theta_n}, e^{-i\theta_1}, \dots, e^{-i\theta_n})Q$$

Now, we make a slight modification and define

$$U(t) = Q^{-1} \text{diag}(e^{i(1-t)\theta_1}, \dots, e^{i(1-t)\theta_n}, e^{-i(1-t)\theta_1}, \dots, e^{-i(1-t)\theta_n})Q$$

for $0 \leq t \leq 1$. $U(t)$ is continuous, $U(0) = U$, and $U(1) = Q^{-1}IQ = I$. By applying the converse of Theorem 1.6 to $U(t)$, it is clear that $U(t) \in Sp(n)$ for all $0 \leq t \leq 1$.

Hence $Sp(n)$ is connected.

For any $U \in Sp(n)$,

$$\begin{aligned} \det(U) &= \det(Q^{-1} \text{diag}(e^{i\theta_1}, \dots, e^{i\theta_n}, e^{-i\theta_1}, \dots, e^{-i\theta_n})Q) \\ &= \det(Q^{-1}) \det(\text{diag}(e^{i\theta_1}, \dots, e^{i\theta_n}, e^{-i\theta_1}, \dots, e^{-i\theta_n})) \det(Q) \\ &= \frac{1}{\det(Q)} e^{i\theta_1} \dots e^{i\theta_n} e^{-i\theta_1} \dots e^{-i\theta_n} \det(Q) \\ &= 1 \end{aligned}$$

□

Exercise 1.9 Suppose a is an irrational real number. Show that the set E_a of numbers of the form $e^{2\pi i n a}$, $n \in \mathbb{Z}$, is dense in the unit circle S^1 .

Proof. We follow the given hint, and divide S^1 into N equally sized bins of length $\frac{2\pi}{N}$. E_a has an infinite number of elements, and N may be large but finite. There must be at least one bin that has an infinite number of elements of E_a (otherwise, if every bin has finite number of elements and N is finite, then E_a would be finite).

Now, choose a bin with infinitely many elements of E_a . In this bin, it is possible to choose two distinct elements of E_a of the form $e^{2\pi i n a}$ and $e^{2\pi i m a}$, with $n, m \in \mathbb{R}$, $n \neq m$.

We use the fact that E_a is a subgroup of S^1 . $e^{2\pi i n a}, e^{2\pi i m a} \in E_a$, so the inverse $e^{-2\pi i m a} \in E_a$, so $e^{2\pi i (n-m)a} \in E_a$, and so $(e^{2\pi i (n-m)a})^k = e^{2k\pi i (n-m)a} \in E_a$ for any $k \in \mathbb{Z}$.

By taking N to be large, we can force $e^{2\pi i n a}$ and $e^{2\pi i m a}$ to be arbitrarily close to each other on S^1 . Then $e^{2k\pi i (n-m)a}$ is a rotation by an arbitrarily small angle. Putting it to the power of k allows us to approximate any point on S^1 by an element of E_a .

More formally, let $e^{i\theta} \in S^1$ for any $\theta \in \mathbb{R}$.

$$\begin{aligned} |e^{i\theta} - e^{2k\pi i (n-m)a}| &= \left| e^{2k\pi i (n-m)a} \left(\frac{e^{i\theta}}{e^{2k\pi i (n-m)a}} - 1 \right) \right| \\ &= \left| \frac{e^{i\theta}}{e^{2k\pi i (n-m)a}} - 1 \right| \\ &= |e^{i(\theta - 2k\pi(n-m)a)} - 1| \\ &= |e^{i(\theta - 2k\pi(n-m)a)} - e^0| \end{aligned}$$

By continuity, $\forall \varepsilon > 0$, $\exists \delta > 0$ such that $|\theta - 2k\pi(n-m)a| < \delta \Rightarrow |e^{i(\theta - 2k\pi(n-m)a)} - e^0| < \varepsilon$. So $\forall \varepsilon > 0$, we can choose N large, and a suitable k such that

$$|e^{i\theta} - e^{2k\pi i (n-m)a}| < \varepsilon$$

Hence E_a is dense in S^1 . □

2 The Matrix Exponential

2.1 The Exponential of a Matrix

If X is a $n \times n$ matrix, we define the **exponential** of X , denoted e^X , by the power series

$$e^X = \sum_{m=0}^{\infty} \frac{X^m}{m!} \quad (2.1)$$

where X^0 is the identity matrix I and X^m is the repeated matrix product of X with itself.

Proposition 2.1. *The series 2.1 converges for all $X \in M_n(\mathbb{C})$ and e^X is a continuous function of X .*

Definition 2.2. *For any $X \in M_n(\mathbb{C})$, we define the **Hilbert-Schmidt** norm of X ,*

$$\|X\| = \left(\sum_{j,k=1}^n |X_{jk}|^2 \right)^{1/2}$$

The Hilbert-Schmidt norm may be computed in a basis-independent way as

$$\|X\| = (\text{tr}(X^*X))^{1/2}$$

This norm satisfies

$$\|X + Y\| \leq \|X\| + \|Y\|$$

$$\|XY\| \leq \|X\| \|Y\|$$

for all $X, Y \in M_n(\mathbb{C})$.

Proposition 2.3. *Let X and Y be arbitrary $n \times n$ matrices. Then we have the following:*

1. $e^0 = I$
2. $(e^X)^* = e^{X^*}$
3. e^X is invertible and $(e^X)^{-1} = e^{-X}$
4. $e^{(\alpha+\beta)X} = e^{\alpha X} e^{\beta X}$ for all α and β in $\mathbb{C}$
5. If $XY = YX$, then $e^{X+Y} = e^X e^Y = e^Y e^X$
6. If $C \in GL(N, \mathbb{C})$, then $e^{CXC^{-1}} = Ce^XC^{-1}$

Proposition 2.4. *Let X be a $n \times n$ complex matrix. Then e^{tX} is a smooth curve in $M_n(\mathbb{C})$ and*

$$\frac{d}{dt} e^{tX} = X e^{tX} = e^{tX} X$$

In particular,

$$\left. \frac{d}{dt} e^{tX} \right|_{t=0} = X$$

2.2 Computing the Exponential

The book explains some methods for computing the matrix exponential. If X is diagonalizable, then $X = CDC^{-1}$, and $e^X = Ce^DC^{-1}$. If X is nilpotent, ($X^k = 0$ for some k), then the power series of e^X terminates. Finally, every matrix X can be written uniquely as $X = S + N$ where S is diagonalizable, N is nilpotent, and $SN = NS$. Then $e^X = e^{S+N} = e^S e^N$.

2.3 The Matrix Logarithm

We want to define a matrix logarithm, which should be an inverse function to the matrix exponential. The simplest way is by a power series. Recall that in the case of complex numbers,

Lemma 2.5. *The function*

$$\log z = \sum_{m=1}^{\infty} (-1)^{m+1} \frac{(z-1)^m}{m}$$

is defined and holomorphic in a circle of radius 1 about $z = 1$. For all z with $|z - 1| < 1$,

$$e^{\log z} = z$$

For all u with $|u| < \log 2$, we have $|e^u - 1| < 1$ and

$$\log e^u = u$$

Definition 2.6. *For an $n \times n$ matrix A , define $\log A$ by*

$$\log A = \sum_{m=1}^{\infty} (-1)^{m+1} \frac{(A - I)^m}{m}$$

whenever the series converges.

Theorem 2.7. *The function*

$$\log A = \sum_{m=1}^{\infty} (-1)^{m+1} \frac{(A - I)^m}{m}$$

is defined and continuous on the set of all $n \times n$ complex matrices A with $\|A - I\| < 1$. For all $A \in M_n(\mathbb{C})$ with $\|A - I\| < 1$,

$$e^{\log A} = A$$

For all $X \in M_n(\mathbb{C})$ with $\|X\| < \log 2$, $\|e^X - I\| < 1$ and

$$\log e^X = X$$

Notice from the last sentence that it is not true that $\log e^X = X$ whenever the series for the logarithm converges.

Proposition 2.8. *There exists a constant c such that for all $n \times n$ matrices B with $\|B\| < 1/2$, we have*

$$\|\log(I + B) - B\| \leq c\|B\|^2$$

More concisely, the proposition can be stated as

$$\log(I + B) = B + O(\|B\|^2)$$

Theorem 2.9. *Every invertible $n \times n$ matrix can be expressed as e^X for some $X \in M_n(\mathbb{C})$.*

2.4 Further Properties of the Exponential

Theorem 2.10 (Lie Product Formula). *For all $X, Y \in M_n(\mathbb{C})$, we have*

$$e^{X+Y} = \lim_{m \rightarrow \infty} \left(e^{\frac{X}{m}} e^{\frac{Y}{m}} \right)^m$$

Theorem 2.11. *For any $X \in M_n(\mathbb{C})$, we have*

$$\det(e^X) = e^{\text{tr}(X)}$$

Definition 2.12. *A function $A : \mathbb{R} \rightarrow GL(n, \mathbb{C})$ is called a **one-parameter subgroup** of $GL(n, \mathbb{C})$ if*

1. A is continuous,
2. $A(0) = I$,
3. $A(t+s) = A(t)A(s)$ for all $t, s \in \mathbb{R}$.

Theorem 2.13 (One-Parameter Subgroups). *If $A(\cdot)$ is a one-parameter subgroup of $GL(n, \mathbb{C})$, there exists a unique $n \times n$ complex matrix X such that*

$$A(t) = e^{tX}$$

Lemma 2.14. *Fix some ε with $\varepsilon < \log 2$. Let $B_{\varepsilon/2}$ be the ball of radius $\varepsilon/2$ around the origin in $M_n(\mathbb{C})$, and let $U = \exp(B_{\varepsilon/2})$. Then every $B \in U$ has a unique square root C in U , given by $C = \exp(\frac{1}{2} \log B)$.*

Proposition 2.15. *The exponential map is an infinitely differentiable map of $M_n(\mathbb{C})$ into $M_n(\mathbb{C})$.*

2.5 The Polar Decomposition

The polar decomposition for a nonzero complex number z states that z can be written uniquely as $z = up$, where $|u| = 1$ and p is real and positive. Since p is real and positive, it can be written as $p = e^x$ for a unique real number x . This gives the polar decomposition for z ,

$$z = ue^x$$

with $x \in \mathbb{R}$ and $|u| = 1$.

We want to establish a similar polar decomposition for $GL(n, \mathbb{C})$, then for various subgroups. If P is a self-adjoint $n \times n$ matrix ($P^* = P$), we say that P is **positive** if $\langle v, Pv \rangle > 0$ for all nonzero $v \in \mathbb{C}^n$. A self-adjoint matrix P is positive if and only if all the eigenvalues of P are positive.

Theorem 2.16. *Polar decomposition for $GL(n, \mathbb{C})$.*

1. Every $A \in GL(n, \mathbb{C})$ can be written uniquely in the form

$$A = UP$$

where U is unitary and P is self-adjoint and positive.

2. Every self-adjoint positive matrix P can be written uniquely in the form

$$P = e^X$$

with X self-adjoint. Conversely, if X is self-adjoint, then e^X is self-adjoint and positive.

3. If we decompose each $A \in GL(n, \mathbb{C})$ (uniquely) as

$$A = Ue^X$$

with U unitary and X self-adjoint, then U and X depend continuously on A .

Lemma 2.17. *If Q is a self-adjoint, positive matrix, then Q has a unique self-adjoint, positive square root.*

Proof. Recall that a matrix Q is self-adjoint (Hermitian) if $Q^* = Q$, and unitary if $Q^* = Q^{-1}$. From Appendix A, we know that a matrix Q self-adjoint $\Rightarrow$ normal $\Rightarrow$ diagonalizable, and it is possible to find an orthonormal basis of eigenvectors for Q . Here, we can write $Q = UDU^{-1}$ where U can be taken to be unitary.

Let Q be a self-adjoint, positive matrix, then Q can be written as

$$Q = U \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix} U^{-1}$$

with U unitary. The λ_j are eigenvalues corresponding to the orthonormal basis of eigenvectors of Q . Since Q is self-adjoint and positive, $\lambda_j > 0$ for all j .

Thus, we can construct a square root of Q as

$$Q^{1/2} = U \begin{pmatrix} \lambda_1^{1/2} & & \\ & \ddots & \\ & & \lambda_n^{1/2} \end{pmatrix} U^{-1}$$

$Q^{1/2}$ is still self-adjoint and positive. This proves the existence of the square root.

Consider P a self-adjoint, positive matrix with eigenvalues λ_j and eigenvectors v_j , and so

$$Pv_j = \lambda_j v_j$$

Notice that the eigenspaces of P^2 are the same as the eigenspaces of P , with the eigenvalues of P^2 equal to the squares of the eigenvalues of P . This is because

$$P^2 v_j = P(Pv_j) = P(\lambda_j v_j) = \lambda_j (\lambda_j v_j) = \lambda_j^2 v_j$$

The mapping $x \mapsto x^2$ is injective for $x \in \mathbb{R}^{\geq 0}$. So eigenspaces with distinct eigenvalues remain with distinct eigenvalues after squaring.

If a self-adjoint positive Q has a self-adjoint positive square root P , then the eigenspaces of P must be the same as the eigenspaces of Q , and the eigenvalues of P must be the positive square roots of the eigenvalues of Q . Thus, P is uniquely determined by Q , proving uniqueness. $\square$

Proof. of Theorem 2.16.

1. Note that if $A = UP$, then $A^*A = P^*U^*UP = PU^{-1}UP = P^2$. For any matrix A , the matrix A^*A is self-adjoint, since $(A^*A)^* = A^*A^{**} = A^*A$. If, in addition, A is invertible, then for all nonzero $v \in \mathbb{C}^n$, we have

$$\langle v, A^*Av \rangle = \langle Av, Av \rangle > 0$$

where the last inequality is due to the definition of the inner product and v being nonzero. So A^*A is positive. For all invertible A , let us define P by

$$P = (A^*A)^{1/2}$$

where the square root is the unique self-adjoint positive square root from Lemma 2.17. Then, we define

$$U = AP^{-1} = A[(A^*A)^{1/2}]^{-1}$$

P is self-adjoint and positive by construction, and $A = UP$ by the definition of U , so it remains only to check that U is unitary. To do this, we check

$$U^*U = [(A^*A)^{1/2}]^{-1}A^*A[(A^*A)^{1/2}]^{-1}$$

noting that the inverse of a self-adjoint positive matrix is self-adjoint. Since A^*A is the square of $(A^*A)^{1/2}$, we have

$$U^*U = [(A^*A)^{1/2}]^{-1}(A^*A)^{1/2}(A^*A)^{1/2}[(A^*A)^{1/2}]^{-1} = I$$

showing that U is unitary. This proves existence.

We already showed that if $A = UP$, then $A^*A = P^2$ where A^*A is self-adjoint positive. The uniqueness of P follows from Lemma 2.17. The uniqueness of U then follows from the definition $U = AP^{-1}$.

2. The existence and uniqueness of the decomposition is proven in the same way as in Lemma 2.17. Instead of the square root function, we use the logarithm function, which is a bijection between $(0, \infty)$ and $\mathbb{R}$.

Let P be a self-adjoint matrix, so it is normal, so it is diagonalizable, and so P can be written as

$$P = U \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix} U^{-1}$$

where U can be taken to be unitary, and λ_j is an orthonormal basis of P . Since P is positive, $\lambda_j > 0$ for all j . Next, we define

$$\log P = U \begin{pmatrix} \log \lambda_1 & & \\ & \ddots & \\ & & \log \lambda_n \end{pmatrix} U^{-1} =: X$$

Then,

$$P = \exp \left(U \begin{pmatrix} \log \lambda_1 & & \\ & \ddots & \\ & & \log \lambda_n \end{pmatrix} U^{-1} \right) = e^X$$

We can check that X is self-adjoint

$$\begin{aligned} X^* &= (U^{-1})^* \text{diag}(\log \lambda_1, \dots, \log \lambda_n)^* U^* \\ &= (U^*)^{-1} \text{diag}(\log \lambda_1, \dots, \log \lambda_n)^* U^* \\ &= U \text{diag}(\log \lambda_1, \dots, \log \lambda_n) U^{-1} \\ &= X \end{aligned}$$

So P can be written as e^X where X is self-adjoint, which proves existence.

Moreover, the mapping $P \mapsto \log P$ is a bijection, so every P corresponds to a unique X such that $P = e^X$, which proves uniqueness.

For a self-adjoint matrix X , we know from before that we can write

$$X = U \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix} U^{-1}$$

with U unitary, and λ_j an orthonormal basis of X . Define

$$e^X = U \begin{pmatrix} e_1^\lambda & & \\ & \ddots & \\ & & e_n^\lambda \end{pmatrix} U^{-1}$$

e^X is self-adjoint because

$$\begin{aligned} (e^X)^* &= (U^{-1})^* \text{diag}(e_1^\lambda, \dots, e_n^\lambda)^* U^* \\ &= (U^*)^{-1} \text{diag}(e_1^\lambda, \dots, e_n^\lambda)^* U^* \\ &= U \text{diag}(e_1^\lambda, \dots, e_n^\lambda) U^{-1} \\ &= e^X \end{aligned}$$

Furthermore, e^X is positive because

$$\begin{aligned} \langle v, e^X v \rangle &= \langle v, U \text{diag}(e_1^\lambda, \dots, e_n^\lambda) U^{-1} v \rangle \\ &= \langle U w, U \text{diag}(e_1^\lambda, \dots, e_n^\lambda) w \rangle \\ &= \langle w, U^* U \text{diag}(e_1^\lambda, \dots, e_n^\lambda) w \rangle \\ &= \langle w, \text{diag}(e_1^\lambda, \dots, e_n^\lambda) w \rangle \\ &= w^t \text{diag}(e_1^\lambda, \dots, e_n^\lambda) w \\ &= (w_1 \ \dots \ w_n) \begin{pmatrix} w_1 \\ \vdots \\ w_n \end{pmatrix} \\ &= \sum_{j=1}^n e^{\lambda_j} w_j^2 > 0 \end{aligned}$$

Hence for any self-adjoint X self-adjoint, e^X is self-adjoint and positive.

X is referred to as the "logarithm" of P .

3. We first show that the logarithm X of a self-adjoint positive matrix P depends continuously on P . First, we consider the case where the eigenvalues of P are between 0 and 2. The proof of Theorem 2.7 establishes that if A is diagonalizable and all eigenvalues λ of A satisfy $|\lambda - 1| < 1$, then $\log A$ is defined and $e^{\log A} = A$. Similarly if X is diagonalizable and all eigenvalues λ of X satisfy $|\lambda| < \log 2$, then $\log e^X$ is defined and $\log e^X = X$. So the power series for $\log P$ converges to X .

$$\log P = \sum_{m=1}^{\infty} (-1)^{m+1} \frac{(P - I)^m}{m}$$

is defined for $\|P - I\| < 1$. $(P - I)^m$ is a continuous function of P , so the partial sum is continuous. Note that

$$\left| (-1)^{m+1} \frac{(P - I)^m}{m} \right| = \frac{\|(P - I)^m\|}{m} \leq \frac{\|P - I\|^m}{m} \leq \frac{R^m}{m}, \quad R < 1$$

and that

$$\sum_{m=0}^{\infty} \frac{R^m}{m}$$

converges since

$$\lim_{m \rightarrow \infty} \left| \frac{R^{m+1}}{m+1} \frac{m}{R^m} \right| = \lim_{m \rightarrow \infty} R \frac{m}{m+1} = R < 1$$

By the Weierstrass M-test, the series converges uniformly on all sets of the form $\{\|P - I\| \leq R\}$, for $R < 1$. So $\log P$ is continuous for all $P \in M_n(\mathbb{C})$ such that $\|P - I\| < 1$.

Next, we consider the general case. Fix some self-adjoint positive matrix P_0 . Fix a small neighbourhood $\mathcal{U}$ of P_0 , choose $a > 0$ large. For $P \in \mathcal{U}$, write

$$P = e^a(e^{-a}P) = e^a I(e^{-a}P) = e^{aI}(e^{-a}P)$$

It is easy to see that e^{aI} and $e^{-a}P$ are commutative, and so

$$\log P = \log(e^{aI}(e^{-a}P)) = aI + \log(e^{-a}P)$$

If a is large enough, then for any $P \in \mathcal{U}$, the positive eigenvalues of $e^{-a}P$ all become less than 2, so the series for $\log(e^{-a}P)$ converge and depends continuously on P , thus X depends continuously on P .

By writing

$$P^{1/2} = \exp(\log(P)/2)$$

we see that the square root operation on P is also continuous. The mapping $A \mapsto A^*A$ is continuous, and the mapping $A^*A \mapsto (A^*A)^{1/2}$ is continuous. Since $P = (A^*A)^{1/2}$, P is continuous on A , and so X is continuous on A . Since $U = AP^{-1}$, U is continuous on A .

□

Proposition 2.18. *Polar decomposition for various subgroups of $GL(n, \mathbb{C})$.*

1. Every $A \in GL(n, \mathbb{R})$ can be written uniquely as

$$A = Re^X$$

where $R \in O(n)$ and X is real and symmetric.

2. Every $A \in SL(n, \mathbb{C})$ can be written uniquely as

$$A = Ue^X$$

where $U \in SU(n)$ and X is self-adjoint with trace zero.

3. Every $A \in SL(n, \mathbb{R})$ can be written uniquely as

$$A = Re^X$$

where $R \in SO(n)$ and X is real and symmetric with trace zero.

2.6 Exercises

Exercise 2.6 Show that every 2×2 matrix X with $\text{tr}(X) = 0$ satisfies

$$X^2 = -\det(X)I$$

If X is 2×2 with trace zero, show by direct calculation using the power series for the exponential that

$$e^X = \cos(\sqrt{\det X})I + \frac{\sin \sqrt{\det X}}{\sqrt{\det X}}X$$

where $\sqrt{\det X}$ is either of the two (possibly complex) square roots of $\det X$. Use this to give an alternative computation of the exponential e^{X_1} in Example 2.5.

Proof. Let X be a 2×2 matrix with $\text{tr}(X) = 0$, taking the form

$$X = \begin{pmatrix} a & b \\ c & -a \end{pmatrix}$$

The determinant is

$$\det(X) = -a^2 - bc$$

Hence, we have

$$X^2 = \begin{pmatrix} a & b \\ c & -a \end{pmatrix} \begin{pmatrix} a & b \\ c & -a \end{pmatrix} = \begin{pmatrix} a^2 + bc & 0 \\ 0 & a^2 + bc \end{pmatrix} = (a^2 + bc)I = -\det(X)I$$

which proves the required statement.

The power series of the matrix exponential is

$$e^X = \sum_{n=0}^{\infty} \frac{X^n}{n!} = I + X + \frac{X^2}{2!} + \frac{X^3}{3!} + \frac{X^4}{4!} + \cdots$$

Using the result we just proved, $X^2 = -\det(X)I$, we can generalize and say that

$$X^{2n} = (-1)^n \det(X)^n I$$

$$X^{2n+1} = X^{2n} X = (-1)^n \det(X)^n X$$

The power series of the matrix exponential is absolutely convergent, so we can rearrange terms without worrying about the series converging to some other limit (source).

$$\begin{aligned} e^X &= I + X + \frac{X^2}{2!} + \frac{X^3}{3!} + \frac{X^4}{4!} + \cdots \\ &= \left(I + \frac{X^2}{2!} + \frac{X^4}{4!} + \cdots \right) + \left(X + \frac{X^3}{3!} + \frac{X^5}{5!} + \cdots \right) \\ &= \left(\sum_{n=0}^{\infty} \frac{(-1)^n \det(X)^n}{(2n)!} \right) I + \left(\sum_{n=0}^{\infty} \frac{(-1)^n \det(X)^n}{(2n+1)!} \right) X \\ &= \left(\sum_{n=0}^{\infty} \frac{(-1)^n \sqrt{\det X}^{2n}}{(2n)!} \right) I + \left(\sum_{n=0}^{\infty} \frac{(-1)^n \sqrt{\det X}^{2n+1}}{\sqrt{\det X} (2n+1)!} \right) X \\ &= \cos(\sqrt{\det X}) I + \frac{\sin \sqrt{\det X}}{\sqrt{\det X}} X \end{aligned}$$

where the last equality is from the power series of $\cos$ and $\sin$. This proves the desired identity.

In Example 2.5, the matrix X_1 is

$$X_1 = \begin{pmatrix} 0 & -a \\ a & 0 \end{pmatrix}$$

which is a 2×2 matrix with trace zero. Its determinant is $\det X_1 = a^2$. We apply the above formula to compute its matrix exponential.

$$\begin{aligned} e^{X_1} &= \cos(\sqrt{a^2}) I + \frac{\sin \sqrt{a^2}}{\sqrt{a^2}} X_1 \\ &= \begin{pmatrix} \cos \sqrt{a^2} & 0 \\ 0 & \cos \sqrt{a^2} \end{pmatrix} + \begin{pmatrix} 0 & -a \frac{\sin \sqrt{a^2}}{\sqrt{a^2}} \\ a \frac{\sin \sqrt{a^2}}{\sqrt{a^2}} & 0 \end{pmatrix} \\ &= \begin{pmatrix} \cos a & -\sin a \\ \sin a & \cos a \end{pmatrix} \end{aligned}$$

This is the same result as in Example 2.5. Note that $\sqrt{\det X_1} = \sqrt{a^2}$ can be taken to be either $+a$ or $-a$. The final result is the same. $\square$

3 Lie Algebras

3.1 Definitions and First Examples

Definition 3.1. A *finite-dimensional real or complex Lie algebra* is a finite-dimensional real or complex vector space $\mathfrak{g}$, together with a map $[\cdot, \cdot]$ from $\mathfrak{g} \times \mathfrak{g}$ into $\mathfrak{g}$, with the following properties:

1. $[\cdot, \cdot]$ is bilinear.
2. $[\cdot, \cdot]$ is skew symmetric: $[X, Y] = -[Y, X]$ for all $X, Y \in \mathfrak{g}$.
3. The Jacobi identity holds:

$$[X, [Y, Z]] + [Y, [Z, X]] + [Z, [X, Y]] = 0$$

for all $X, Y, Z \in \mathfrak{g}$.

Two elements X and Y of $\mathfrak{g}$ commute if $[X, Y] = 0$. $\mathfrak{g}$ is commutative if $[X, Y] = 0$ for all $X, Y \in \mathfrak{g}$.

Example 3.2. Let $\mathfrak{g} = \mathbb{R}^3$, let $[\cdot, \cdot] : \mathbb{R}^3 \times \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be given by

$$[x, y] = x \times y$$

where $x \times y$ is the cross product. Then $\mathfrak{g}$ is a Lie algebra.

Example 3.3. Let $\mathcal{A}$ be an associative algebra and let $\mathfrak{g}$ be a subspace of $\mathcal{A}$ such that $XY - YX \in \mathfrak{g}$ for all $X, Y \in \mathfrak{g}$. Then $\mathfrak{g}$ is a Lie algebra with bracket operation given by $[X, Y] = XY - YX$.

Example 3.4. Let $\mathfrak{sl}(n, \mathbb{C})$ denote the space of all $X \in M_n(\mathbb{C})$ for which $\text{tr}(X) = 0$. Then $\mathfrak{sl}(n, \mathbb{C})$ is a Lie algebra with bracket $[X, Y] = XY - YX$.

Definition 3.5. A *subalgebra* of a real or complex Lie algebra $\mathfrak{g}$ is a subspace $\mathfrak{h}$ of $\mathfrak{g}$ such that $[H_1, H_2] \in \mathfrak{h}$ for all $H_1, H_2 \in \mathfrak{h}$. If $\mathfrak{g}$ is a complex Lie algebra and $\mathfrak{h}$ is a real subspace of $\mathfrak{g}$ which is closed under brackets, then $\mathfrak{h}$ is said to be a **real subalgebra** of $\mathfrak{g}$.

A subalgebra $\mathfrak{h}$ of a Lie algebra $\mathfrak{g}$ is said to be an **ideal** in $\mathfrak{g}$ if $[X, H] \in \mathfrak{h}$ for all $X \in \mathfrak{g}$ and $H \in \mathfrak{h}$.

The **center** of a Lie algebra $\mathfrak{g}$ is

$$\mathfrak{z}(\mathfrak{g}) = \{X \in \mathfrak{g} : [X, Y] = 0 \ \forall Y \in \mathfrak{g}\}$$

Example. The vector space $\mathfrak{gl}(n, \mathbb{C})$ of all $n \times n$ complex matrices is a Lie algebra with Lie bracket $[X, Y] = XY - YX$. The subspace $\mathfrak{sl}(n, \mathbb{C})$ of all $n \times n$ complex matrices with trace zero is a Lie subalgebra of $\mathfrak{gl}(n, \mathbb{C})$.

Example. Every Lie algebra $\mathfrak{g}$ has at least two ideals: $\{0\}$ and $\mathfrak{g}$.

$\mathfrak{sl}(n, \mathbb{C})$ is a non-trivial ideal of $\mathfrak{gl}(n, \mathbb{C})$ for $n > 1$.

For a Lie algebra $\mathfrak{g}$, its center $\mathfrak{z}(\mathfrak{g})$ and its derived algebra $\mathcal{D}(\mathfrak{g}) = [\mathfrak{g}, \mathfrak{g}]$ are ideals.

Example. The center of the Heisenberg Lie algebra is

$$\mathfrak{z}(\mathfrak{h}) = \left\{ \begin{pmatrix} 0 & 0 & c \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} : c \in \mathbb{R} \right\} = \{cZ : c \in \mathbb{R}\}$$

Definition 3.6. If $\mathfrak{g}$ and $\mathfrak{h}$ are Lie algebras, then a linear map $\phi : \mathfrak{g} \rightarrow \mathfrak{h}$ is called a **Lie algebra homomorphism** if $\phi([X, Y]) = [\phi(X), \phi(Y)]$ for all $X, Y \in \mathfrak{g}$. If, in addition, ϕ is one-to-one and onto (i.e. bijective), then ϕ is called a **Lie algebra isomorphism**. A Lie algebra isomorphism of a Lie algebra with itself is called a **Lie algebra automorphism**.

Definition 3.7. If $\mathfrak{g}$ is a Lie algebra and X is an element of $\mathfrak{g}$, define a linear map $\text{ad}_X : \mathfrak{g} \rightarrow \mathfrak{g}$ by

$$\text{ad}_X(Y) = [X, Y].$$

The map $X \mapsto \text{ad}_X$ is the **adjoint map** or **adjoint representation**.

We can view ad (the map $X \mapsto \text{ad}_X$) as a linear map of $\mathfrak{g}$ into $\text{End}(\mathfrak{g})$, the space of linear operators on $\mathfrak{g}$.

Proposition 3.8. If $\mathfrak{g}$ is a Lie algebra, then

$$\text{ad}_{[X, Y]} = \text{ad}_X \text{ad}_Y - \text{ad}_Y \text{ad}_X = [\text{ad}_X, \text{ad}_Y];$$

that is, $\text{ad} : \mathfrak{g} \mapsto \text{End}(\mathfrak{g})$ is a Lie algebra homomorphism.

Example. If $\mathfrak{g}$ is a Lie algebra, then the identity map $x \mapsto x$ is a Lie algebra isomorphism, and also a Lie algebra automorphism.

Let K be a field and $\mathfrak{gl}(n, K)$ be the Lie algebra given by all $n \times n$ matrices over K , with Lie bracket given by $[X, Y] = XY - YX$. For $X \in \mathfrak{gl}(K)$ and $g \in GL(n, K)$, the map $X \mapsto gXg^{-1}$ is an isomorphism from $\mathfrak{gl}(n, K)$ to itself, i.e. an automorphism of $\mathfrak{gl}(n, K)$.

Definition 3.9. If $\mathfrak{g}_1$ and $\mathfrak{g}_2$ are Lie algebras, the **direct sum** of $\mathfrak{g}_1$ and $\mathfrak{g}_2$ is the vector space direct sum of $\mathfrak{g}_1$ and $\mathfrak{g}_2$, with bracket given by

$$[(X_1, X_2), (Y_1, Y_2)] = ([X_1, Y_1], [X_2, Y_2]).$$

If $\mathfrak{g}$ is a Lie algebra and $\mathfrak{g}_1$ and $\mathfrak{g}_2$ are subalgebras, we say that $\mathfrak{g}$ **decomposes as the Lie algebra direct sum** of $\mathfrak{g}_1$ and $\mathfrak{g}_2$ if $\mathfrak{g}$ is the direct sum of $\mathfrak{g}_1$ and $\mathfrak{g}_2$ as vector spaces and $[X_1, X_2] = 0$ for all $X_1 \in \mathfrak{g}_1$ and $X_2 \in \mathfrak{g}_2$.

Example. Examples of Lie algebra direct sums

- $\mathfrak{g} = \mathbb{R} \oplus \mathbb{R}$
- $\mathfrak{gl}(2, \mathbb{C}) = \mathfrak{sl}(2, \mathbb{C}) \oplus \mathbb{C}I$

Definition 3.10. Let $\mathfrak{g}$ be a finite-dimensional real or complex Lie algebra, and let $X_1, \dots, X_N$ be a basis for $\mathfrak{g}$ (as a vector space). Then the unique constants c_{jkl} such that

$$[X_j, X_k] = \sum_{l=1}^N c_{jkl} X_l$$

are called the **structure constants** of $\mathfrak{g}$ (with respect to the chosen basis).

The structure constants satisfy the following two conditions:

$$c_{jkl} + c_{kjl} = 0$$

$$\sum_n (c_{jkn} c_{nlm} + c_{kln} c_{njm} + c_{ljn} c_{nkm}) = 0$$

for all j, k, l, m .

Example. A basis for $\mathfrak{sl}(2, \mathbb{C})$ is

$$X_1 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}; \quad X_2 = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}; \quad X_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

and we have the relations

$$[X_3, X_1] = 2X_1; \quad [X_3, X_2] = -2X_2; \quad [X_1, X_2] = X_3$$

The structure constants are

$$c_{311} = 2, \quad c_{322} = -2, \quad c_{123} = 1$$

with all others following from skew symmetry or equal to 0.

3.2 Simple, Solvable, and Nilpotent Lie Algebras

Definition 3.11. A Lie algebra $\mathfrak{g}$ is called **irreducible** if the only ideals in $\mathfrak{g}$ are $\mathfrak{g}$ and $\{0\}$. A Lie algebra $\mathfrak{g}$ is called **simple** if it is irreducible and $\dim \mathfrak{g} \geq 2$

Example. Irreducible but not simple Lie algebras include the one-dimensional Lie algebras. For instance, take $\mathfrak{g} = \mathbb{R}$ with bracket $[x, y] = 0$.

Proposition 3.12. The Lie algebra $\mathfrak{sl}(2, \mathbb{C})$ is simple.

Definition 3.13. If $\mathfrak{g}$ is a Lie algebra, then the **commutator ideal** in $\mathfrak{g}$, denoted $[\mathfrak{g}, \mathfrak{g}]$, is the space of linear combinations of commutators, that is, the space of elements Z in $\mathfrak{g}$ that can be expressed as

$$Z = c_1[X_1, Y_1] + \cdots + c_m[X_m, Y_m]$$

for some constants c_j and vectors $X_j, Y_j \in \mathfrak{g}$.

Example. For $\mathfrak{g} = \mathfrak{gl}(n, \mathbb{C})$, the commutator ideal is $[\mathfrak{g}, \mathfrak{g}] = \mathfrak{sl}(n, \mathbb{C})$.

Definition 3.14. For any Lie algebra $\mathfrak{g}$, we define a sequence of subalgebras $\mathfrak{g}_0, \mathfrak{g}_1, \mathfrak{g}_2, \dots$ of $\mathfrak{g}$ inductively as follows: $\mathfrak{g}_0 = \mathfrak{g}$, $\mathfrak{g}_1 = [\mathfrak{g}_0, \mathfrak{g}_0]$, $\mathfrak{g}_2 = [\mathfrak{g}_1, \mathfrak{g}_1]$, etc. These subalgebras are called the **derived series** of $\mathfrak{g}$. A Lie algebra $\mathfrak{g}$ is called **solvable** if $\mathfrak{g}_j = \{0\}$ for some j .

It can be shown, using the Jacobi identity and induction on j , that each $\mathfrak{g}_j$ is an ideal in $\mathfrak{g}$.

Definition 3.15. For any Lie algebra $\mathfrak{g}$, we define a sequence of ideals $\mathfrak{g}^j$ in $\mathfrak{g}$ inductively as follows. We set $\mathfrak{g}^0 = \mathfrak{g}$ and then define $\mathfrak{g}^{j+1}$ to be the space of linear combinations of commutators of the form $[X, Y]$ with $X \in \mathfrak{g}$ and $Y \in \mathfrak{g}^j$. These algebras are called the **upper central series** of $\mathfrak{g}$. A Lie algebra $\mathfrak{g}$ is said to be **nilpotent** if $\mathfrak{g}^j = \{0\}$ for some j .

Proposition 3.16. If $\mathfrak{g} \subset M_3(\mathbb{R})$ denotes the space of 3×3 upper triangular matrices with zeros on the diagonal, then $\mathfrak{g}$ satisfies the assumptions of Example 3.3. The Lie algebra $\mathfrak{g}$ is a nilpotent Lie algebra.

Proposition 3.17. If $\mathfrak{g} \subset M_2(\mathbb{C})$ denotes the space of 2×2 matrices of the form

$$\begin{pmatrix} a & b \\ 0 & c \end{pmatrix}$$

with $a, b, c \in \mathbb{C}$, then $\mathfrak{g}$ satisfies the assumptions of Example 3.3. The Lie algebra $\mathfrak{g}$ is solvable but not nilpotent.

I will use the next two examples to develop Proposition 3.16 and 3.17.

Example. Take

$$\mathfrak{g} = \left\{ \begin{pmatrix} 0 & a & b \\ 0 & 0 & c \\ 0 & 0 & 0 \end{pmatrix} : a, b, c \in \mathbb{R} \right\}$$

The upper central series is as follows. Set $\mathfrak{g}^0 = \mathfrak{g}$. Then

$$\mathfrak{g}^1 = [\mathfrak{g}^0, \mathfrak{g}^0] = \left\{ \begin{pmatrix} 0 & 0 & b \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} : b \in \mathbb{R} \right\}$$

$$\mathfrak{g}^2 = \{0\}$$

Hence $\mathfrak{g}$ is nilpotent.

Example. Take

$$\mathfrak{g} = \left\{ \begin{pmatrix} a & b \\ 0 & c \end{pmatrix} : a, b, c \in \mathbb{C} \right\}$$

which is actually a subalgebra of $\mathfrak{gl}(2, \mathbb{C})$. The derived series is as follows. Define $\mathfrak{g}_0 = \mathfrak{g}$, then

$$\mathfrak{g}_1 = [\mathfrak{g}_0, \mathfrak{g}_0] = \left\{ \begin{pmatrix} 0 & b \\ 0 & 0 \end{pmatrix} : b \in \mathbb{C} \right\}$$

$$\mathfrak{g}_2 = [\mathfrak{g}_1, \mathfrak{g}_1] = \{0\}$$

Hence $\mathfrak{g}$ is solvable.

3.3 The Lie Algebra of a Matrix Lie Group

3.4 Examples

3.5 Lie Group and Lie Algebra Homomorphisms

3.6 The Complexification of a Real Lie Algebra

3.7 The Exponential Map

3.8 Consequences of Theorem 3.42

3.9 Exercises

Exercise 3.8 Show that two isomorphic matrix Lie groups have isomorphic Lie algebras.

Proof. Let G and H be two isomorphic matrix Lie groups ($G \cong H$) with corresponding Lie algebras $\mathfrak{g}$ and $\mathfrak{h}$, respectively. Since $G \cong H$, let $\Phi : G \rightarrow H$ be a Lie group isomorphism. By Theorem 3.28, there exists a unique real-linear map $\phi : \mathfrak{g} \rightarrow \mathfrak{h}$ such that

$$\Phi(e^X) = e^{\phi(X)}$$

for all $X \in \mathfrak{g}$. The map ϕ has the properties

1. $\phi(AXA^{-1}) = \Phi(A)\phi(X)\Phi(A)^{-1}$ for all $X \in \mathfrak{g}, A \in G$
2. $\phi([X, Y]) = [\phi(X), \phi(Y)]$, for all $X, Y \in \mathfrak{g}$
3. $\phi(X) = \frac{d}{dt}\Phi(e^{tX})|_{t=0}$ for all $X \in \mathfrak{g}$.

From property 2, it is clear that ϕ is a Lie algebra homomorphism. It remains to show that ϕ is bijective.

Since Φ is a Lie group isomorphism, it must be bijective, hence injective. So $\ker(\Phi) = \{e_G\}$. By Proposition 3.31, we have

$$\ker(\phi) = \text{Lie}(\ker(\Phi)) = \text{Lie}(\{e_G\}) = \{0\}$$

where 0 is the zero element of $\mathfrak{g}$. This implies ϕ is injective. I assume that we are only working with finite-dimensional Lie groups and finite-dimensional Lie algebras. In this case, the dimension of a Lie group equals the dimension of its Lie algebra. Also, since $G \cong H$, $\dim G = \dim H$. So we have

$$\dim \mathfrak{g} = \dim G = \dim H = \dim \mathfrak{h}$$

Since ϕ is injective and $\dim \mathfrak{g} = \dim \mathfrak{h}$, ϕ is bijective.

Hence ϕ is an isomorphism, and the Lie algebras $\mathfrak{g}$ and $\mathfrak{h}$ are isomorphic. $\square$

Remark. Based on the discussion during the groupe de travail, to complete the proof, we still need to show that ϕ^{-1} is a Lie group homomorphism, i.e. it satisfies

$$\phi^{-1}([X, Y]) = [\phi^{-1}(X), \phi^{-1}(Y)]$$

To do this, we can apply ϕ to both sides, and see that both yield $[X, Y]$. Since ϕ is injective, the condition has been proven. However, I am not sure why this extra step is needed.

Exercise 3.13 Let G be a matrix Lie group and let $\mathfrak{g}$ be its Lie algebra. For each $A \in G$, show that Ad_A is a Lie algebra automorphism of $\mathfrak{g}$.

Proof. For each $A \in G$,

$$\begin{aligned} [\text{Ad}_A(X), \text{Ad}_A(Y)] &= \text{Ad}_A(X)\text{Ad}_A(Y) - \text{Ad}_A(Y)\text{Ad}_A(X) \\ &= (AXA^{-1})(AYA^{-1}) - (AYA^{-1})(AXA^{-1}) \\ &= A(XY)A^{-1} - A(YX)A^{-1} \\ &= A(XY - YX)A^{-1} \\ &= A[X, Y]A^{-1} \\ &= \text{Ad}_A([X, Y]) \end{aligned}$$

for all $X, Y \in \mathfrak{g}$. Hence $\text{Ad}_A : \mathfrak{g} \rightarrow \mathfrak{g}$ is a Lie algebra homomorphism. It remains to prove that Ad_A is bijective.

Let $X, Y \in \mathfrak{g}$, with $X \neq Y$. We have $\text{Ad}_A(X) = AXA^{-1}$ and $\text{Ad}_A(Y) = AYA^{-1}$. Suppose $\text{Ad}_A(X) = \text{Ad}_A(Y)$, then $AXA^{-1} = AYA^{-1}$, then by left and right multiplication by A , $X = Y$, which is a contradiction. So Ad_A is an injective mapping.

Let $X \in \mathfrak{g}$. Define $P = A^{-1}XA$. By Theorem 3.20, $AXA^{-1} \in \mathfrak{g}$ for all $A \in G$. Since G is a group, $A^{-1} \in G$, so we see that $P = A^{-1}XA \in \mathfrak{g}$ for each $A \in G$. It is clear that

$$\text{Ad}_A(P) = APA^{-1} = X$$

This means $\forall X \in \mathfrak{g}, \exists P \in \mathfrak{g}$ such that $\text{Ad}_A(P) = X$. So Ad_A is a surjective mapping.

Hence $\text{Ad}_A : \mathfrak{g} \rightarrow \mathfrak{g}$ is a Lie algebra isomorphism, and furthermore a Lie algebra automorphism. $\square$

4 Basic Representation Theory

4.1 Representations

Definition 4.1. Let G be a matrix Lie group. A **finite-dimensional complex representation** of G is a Lie group homomorphism

$$\Pi : G \rightarrow GL(V)$$

where V is a finite-dimensional complex vector space (with $\dim(V) \geq 1$). A **finite-dimensional real representation** of G is a Lie group homomorphism Π of G into $GL(V)$ with V is a finite-dimensional real vector space.

If $\mathfrak{g}$ is a real or complex Lie algebra, then a **finite-dimensional complex representation** of $\mathfrak{g}$ is a Lie algebra homomorphism $\pi : \mathfrak{g} \rightarrow gl(V)$, where V is a finite-dimensional complex vector space. If $\mathfrak{g}$ is a real Lie algebra, then a **finite-dimensional real representation** of $\mathfrak{g}$ is a Lie algebra homomorphism $\pi : \mathfrak{g} \rightarrow gl(V)$, where V is a finite-dimensional real vector space.

If Π or π is an injective homomorphism, the representation is called **faithful**.

Definition 4.2. Let Π be a finite-dimensional real or complex representation of a matrix Lie group G , acting on a space V . A subspace W of V is called **invariant** if $\Pi(A)w \in W$ for all $w \in W$ and all $A \in G$. An invariant subspace W is called **nontrivial** if $W \neq \{0\}$ and $W \neq V$. A representation with no nontrivial invariant subspaces is called **irreducible**.

The terms **invariant**, **nontrivial**, and **irreducible** are defined analogously for representations of Lie algebras.

We may consider complex representations of $\mathfrak{g}$.

Definition 4.3. Let G be a matrix Lie group, let Π be a representation of G acting on the space V , and let Σ be a representation of G acting on the space W . A linear map $\phi : V \rightarrow W$ is called an **intertwining map** of representations if

$$\phi(\Pi(A)v) = \Sigma(A)\phi(v)$$

for all $A \in G$ and all $v \in V$. The analogous property defines intertwining maps of representations of a Lie algebra.

The following proposition is a result of Theorem 3.28.

Proposition 4.4. Let G be a matrix Lie group with Lie algebra $\mathfrak{g}$ and let Π be a finite-dimensional real or complex representation of G , acting on the space V . Then there is a unique representation π of $\mathfrak{g}$ acting on the same space such that

$$\Pi(e^X) = e^{\pi(X)}$$

for all $X \in \mathfrak{g}$. The representation π can be computed as

$$\pi(X) = \left. \frac{d}{dt} \Pi(e^{tX}) \right|_{t=0}$$

and satisfies

$$\pi(AXA^{-1}) = \Pi(A)\pi(X)\Pi(A)^{-1}$$

for all $X \in \mathfrak{g}$ and all $A \in G$.

Proposition 4.5. 1. Let G be a connected matrix Lie group with Lie algebra $\mathfrak{g}$. Let Π be a representation of G and π the associated representation of $\mathfrak{g}$. Then Π is irreducible if and only if π is irreducible.

2. Let G be a connected matrix Lie group, let Π_1 and Π_2 be representations of G , and let π_1 and π_2 be the associated Lie algebra representations. Then π_1 and π_2 are isomorphic if and only if Π_1 and Π_2 are isomorphic.

Proposition 4.6. Let $\mathfrak{g}$ be a real Lie algebra and $\mathfrak{g}_{\mathbb{C}}$ its complexification. Then every finite-dimensional complex representation π of $\mathfrak{g}$ has a unique extension to a complex-linear representation of $\mathfrak{g}_{\mathbb{C}}$, also denoted π . Furthermore, π is irreducible as a representation of $\mathfrak{g}_{\mathbb{C}}$ if and only if it is irreducible as a representation of $\mathfrak{g}$.

The extension of π to $\mathfrak{g}_{\mathbb{C}}$ is given by $\pi(X + iY) = \pi(X) + i\pi(Y)$ for all $X, Y \in \mathfrak{g}$.

Definition 4.7. If V is a finite-dimensional inner product space and G is a matrix Lie group, a representation $\Pi : G \rightarrow GL(V)$ is **unitary** if $\Pi(A)$ is a unitary operator on V for every $A \in G$.

Proposition 4.8. Suppose G is a matrix Lie group with Lie algebra $\mathfrak{g}$. Suppose V is a finite-dimensional inner product space, Π is a representation of G acting on V , and π is the associated representation of $\mathfrak{g}$. If Π is unitary, then $\pi(X)$ is skew self-adjoint for all $X \in \mathfrak{g}$. Conversely, if G is connected and $\pi(X)$ is skew self-adjoint for all $X \in \mathfrak{g}$, then Π is unitary.

A representation π of a real Lie algebra $\mathfrak{g}$ acting on a finite-dimensional inner product space is **unitary** if $\pi(X)$ is skew self-adjoint for all $X \in \mathfrak{g}$.

4.2 Examples of Representations

A matrix Lie group G is a subset of some $GL(n, \mathbb{C})$. The inclusion map of G into $GL(n, \mathbb{C})$, $\Pi(A) = A$, is called the **standard representation** of G . Similarly, if $\mathfrak{g} \subset M_n(\mathbb{C})$ is a Lie algebra of matrices, the map $\pi(X) = X$ is called the **standard representation** of $\mathfrak{g}$.

For any matrix Lie group G , we can define the **trivial representation** $\Pi : G \rightarrow GL(1, \mathbb{C})$ by

$$\Pi(A) = I$$

for all $A \in G$. This is an irreducible representation. For a Lie algebra $\mathfrak{g}$, we can define the **trivial representation** of $\mathfrak{g}$, $\pi : \mathfrak{g} \rightarrow gl(1, \mathbb{C})$ by

$$\pi(X) = 0$$

for all $X \in \mathfrak{g}$. This is an irreducible representation.

Definition 4.9. If G is a matrix Lie group with Lie algebra $\mathfrak{g}$, the **adjoint representation** of G is the map $Ad : G \rightarrow GL(\mathfrak{g})$ given by $A \mapsto Ad_A$. Similarly, the **adjoint representation** of a finite-dimensional Lie algebra $\mathfrak{g}$ is the map $ad : \mathfrak{g} \rightarrow gl(\mathfrak{g})$ given by $X \mapsto ad_X$.

Example 4.10. Let V_m denote the space of homogeneous polynomials of degree m in two complex variables. For each $U \in SU(2)$, define a linear transformation $\Pi_m(U)$ on the space V_m by the formula

$$[\Pi_m(U)f](z) = f(U^{-1}z), \quad z \in \mathbb{C}^2$$

Then Π_m is a representation of $SU(2)$.

Proposition 4.11. For each $m \geq 0$, the representation π_m is irreducible.

4.3 New Representations from Old

4.3.1 Direct Sums

Definition 4.12. Let G be a matrix Lie group and let $\Pi_1, \Pi_2, \dots, \Pi_m$ be representations of G acting on vector spaces $V_1, V_2, \dots, V_m$. Then the **direct sum** of $\Pi_1, \Pi_2, \dots, \Pi_m$ is a representation $\Pi_1 \oplus \dots \oplus \Pi_m$ of G acting on the space $V_1 \oplus \dots \oplus V_m$, defined by

$$[\Pi_1 \oplus \dots \oplus \Pi_m(A)](v_1, \dots, v_m) = (\Pi_1(A)v_1, \dots, \Pi_m(A)v_m)$$

for all $A \in G$.

Similarly, if $\mathfrak{g}$ is a Lie algebra, and $\pi_1, \pi_2, \dots, \pi_m$ are representations of $\mathfrak{g}$ acting on vector spaces $V_1, V_2, \dots, V_m$, then we define the **direct sum** of $\pi_1, \pi_2, \dots, \pi_m$, acting on $V_1 \oplus \dots \oplus V_m$ by

$$[\pi_1 \oplus \dots \oplus \pi_m(X)](v_1, \dots, v_m) = (\pi_1(X)v_1, \dots, \pi_m(X)v_m)$$

for all $X \in \mathfrak{g}$.

4.3.2 Tensor Products

Definition 4.13. If U and V are finite-dimensional real or complex vector spaces, then a **tensor product** of U with V is a vector space W , together with a bilinear map $\phi : U \times V \rightarrow W$ with the following property: if ψ is any bilinear map of $U \times V$ into a vector space X , there exists a unique linear map $\tilde{\psi}$ of W into X such that the following diagram commutes.

$$\begin{array}{ccc} U \times V & \xrightarrow{\phi} & W \\ \psi \searrow & & \swarrow \tilde{\psi} \\ & X & \end{array}$$

4.3.3 Dual Representations

4.4 Complete Reducibility

4.5 Schur's Lemma

4.6 Representations of $\mathfrak{sl}(2; \mathbb{C})$

4.7 Group Versus Lie Algebra Representations

4.8 A Nonmatrix Lie Group

4.9 Exercises